



IONIC EQUILIBRIUM-I

Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A) : pH calculation : Solutions of polyprotic weak acids, Solutions of Polyacidic weak bases

Commit to memory :

Solution of a polyprotic weak acid (H_xA) : $pH = \frac{1}{2} (pK_{a_1} - \log C)$ [if $\alpha_1 \leq 0.1$ or 10%] ; $[H_{x-2}A^{2-}] = K_{a_2}$

Solution of a polyprotic weak base (B) : $pOH = \frac{1}{2} (pK_{b_1} - \log C)$ [if $\alpha_1 \leq 0.1$ or 10%] ; $[BH_2^{2+}] = K_{b_2}$

A-1. What is the pH of 0.01 M H_2S solution ? Also determine $[HS^-]$ & $[S^{2-}]$. Given: For H_2S , $K_{a_1} = 9 \times 10^{-8}$, $K_{a_2} = 1.2 \times 10^{-13}$.

A-2. Calculate the pH of 0.74 g/L aqueous solution of propane-1,2-diamine. Given : pK_{b_1} & pK_{b_2} for base is 4.18 & 7.39 respectively.

Section (B) : pH calculation : Solutions containing mixture of weak monoprotic acid and strong acid, Solutions containing mixture of weak monoacidic base and strong base

Commit to memory :

Mixture of weak acid (monoprotic) and a strong acid: $[H^+]_f = C_1$ (just not considering the H^+ ions from WA).

Mixture of weak base (monoacidic) and a strong base: $[OH^-]_f = C_1$ (just not considering the H^+ ions from WB).

B-1. Upon passing 0.01 mole HCl gas through 100 mL of 0.05 M Formic acid solution ($K_a = 1.8 \times 10^{-4}$), determine change in pH of solution and $[HCOO^-]$ in resulting solution.

B-2. Determine pH & $[NH_4^+]$ in a solution obtained by mixing equal volumes of 0.02 M KOH solution and 0.2 M $NH_3 \cdot H_2O$ solution ($K_b = 2 \times 10^{-5}$). Also calculate % dissociation of $NH_3 \cdot H_2O$ in the final solution.

Section (C) : pH calculation : Solutions containing mixture of weak acids, Solutions containing mixture of weak bases

Commit to memory :

Mixture of two weak acids (both monoprotic) : $[H^+] = C_1\alpha_1 + C_2\alpha_2 = \sqrt{C_1K_{a_1} + C_2K_{a_2}}$

Mixture of two weak bases (both monoacidic) : $[OH^-] = C_1\alpha_1 + C_2\alpha_2 = \sqrt{C_1K_{b_1} + C_2K_{b_2}}$

C-1. Calculate $[H^+]$, $[HCOO^-]$ and $[OCN^-]$ in a solution that contains 0.1 M $HCOOH$ ($K_a = 2.4 \times 10^{-4}$) and 0.1 M $HOCl$ ($K_a = 4 \times 10^{-4}$).

Section (D) : pH calculation : Solutions containing mixture of weak polyprotic acid and strong acid, Solutions containing mixture of weak Polyacidic base and strong base

Commit to memory :

Mixture of a polyprotic weak acid and a strong acid : $[H^+] = [H^+]$ from SA

Mixture of a polyprotic weak base and a strong base : $[OH^-] = [OH^-]$ from SB

D-1. What is $[HS^-]$ & $[S^{2-}]$ in solution of 0.01 M HCl and 0.1 M H_2S ? (Given that for H_2S : $K_{a_1} = 10^{-7}$ and $K_{a_2} = 10^{-14}$)


Section (E) : pH calculation : Solutions of salt of weak polyprotic acid and strong base, Solutions of salt of weak Polyacidic base and strong acid, Solutions of amphiprotic species.
Commit to memory :

 Solution containing polyvalent anion (A^{x-}) of weak polyprotic acid (H_xA) :

$$K_{h_1} = \frac{K_w}{K_{a_x}} ; h = \sqrt{\frac{K_{h_1}}{C}} ; pH = \frac{1}{2}[pK_w + pK_{a_x} + \log C]$$

 Solution containing polyvalent cation (BH_y^{y+}) of weak polyprotic base (B) :

$$K_{h_1} = \frac{K_w}{K_{b_y}} ; h = \sqrt{\frac{K_{h_1}}{C}} ; pH = \frac{1}{2}[pK_w - pK_{b_y} - \log C]$$

$$\text{Solution containing amphiprotic anion of acid } H_xA : pH(H_{x-1}A^-) = \left(\frac{pK_{a_1} + pK_{a_2}}{2} \right) ; pH(H_{x-2}A^{2-}) = \left(\frac{pK_{a_2} + pK_{a_3}}{2} \right)$$

- E-1.** Calculate the pH of 0.1 M Na_3A of solution (salt of a tribasic acid H_3A). Assume only first step hydrolysis to be significant. Given : K_{a_1}, K_{a_2} & K_{a_3} for H_3A are $10^{-4}, 10^{-7}$ & 10^{-9} respectively.
- E-2.** What is the pH of M/20 $KHC_8H_4O_4$ solution ? Given : $H_2C_8H_4O_4$ is a dibasic acid with pK_{a_1} & pK_{a_2} as 2.94 & 5.44 respectively.

PART - II : ONLY ONE OPTION CORRECT TYPE
Section (A) : pH calculation : Solutions of polyprotic weak acids, Solutions of Polyacidic weak bases
Commit to memory :

 Solution of a polyprotic weak acid (H_xA) : $pH = \frac{1}{2}(pK_{a_1} - \log C)$ [if $\alpha_1 \leq 0.1$ or 10%] ; $[H_{x-2}A^{2-}] = K_{a_2}$

 Solution of a polyprotic weak base (B) : $pOH = \frac{1}{2}(pK_{b_1} - \log C)$ [if $\alpha_1 \leq 0.1$ or 10%] ; $[BH_2^{2+}] = K_{b_2}$

- A-1.** K_{b_1} of N_2H_4 is 4×10^{-6} . Then, what is the acid dissociation constant of $N_2H_5^+$ and $N_2H_6^{2+}$ respectively ?
 (A) data insufficient, 4×10^{-6} (B) data insufficient, 2.5×10^{-8}
 (C) 2.5×10^{-9} , data insufficient (D) 2.5×10^{-9} , 4×10^{-6}
- A-2.** For ortho phosphoric acid,
 $H_3PO_4(aq) + H_2O(aq) \longrightarrow H_3O^+(aq) + H_2PO_4^-(aq) ; K_{a_1}$
 $H_2PO_4^-(aq) + H_2O(aq) \longrightarrow H_3O^+(aq) + HPO_4^{2-}(aq) ; K_{a_2}$
 $HPO_4^{2-}(aq) + H_2O(aq) \longrightarrow H_3O^+(aq) + PO_4^{3-}(aq) ; K_{a_3}$
 The correct order of K_a values is :
 (A) $K_{a_1} > K_{a_2} < K_{a_3}$ (B) $K_{a_1} < K_{a_2} < K_{a_3}$ (C) $K_{a_1} > K_{a_2} > K_{a_3}$ (D) $K_{a_1} < K_{a_2} > K_{a_3}$
- A-3.** In a solution of 0.1 M H_3PO_4 acid : (Given $K_{a_1} = 10^{-3}, K_{a_2} = 10^{-7}, K_{a_3} = 10^{-12}$)
 (i) Concentration of H_3PO_4 is :
 (A) 0.01 M (B) 0.09 M (C) 0.05 M (D) 0.1 M
 (ii) Concentration of $H_2PO_4^-$ is :
 (A) 0.01 M (B) 0.09 M (C) 0.02 M (D) 0.04 M
 (iii) Concentration of HPO_4^{2-} is :
 (A) 10^{-7} M (B) 10^{-8} M (C) 10^{-3} M (D) 10^{-4} M
 (iv) Concentration of PO_4^{3-} is :
 (A) 10^{-20} M (B) 10^{-17} M (C) 10^{-15} M (D) 10^{-12} M



- (v) pH of solution is :
 (A) 1 (B) 2 (C) 4 (D) 5
- A-4.** Generally, in aqueous solutions of weak polyacidic bases, OH^- ions produced from 2nd ionisation are negligible because of :
 (A) $K_{b_2} \ll K_{b_1}$
 (B) Common ion effect exerted by OH^- ions produced from 1st ionisation.
 (C) Both (A) & (B)
 (D) None of these

Section (B) : pH calculation: Solutions containing mixture of weak monoprotic acid and strong acid, Solutions containing mixture of weak monoacidic base and strong base

Commit to memory :

Mixture of weak acid (monoprotic) and a strong acid : $[\text{H}^+]_f = C_1$ (just not considering the H^+ ions from WA).

Mixture of weak base (monoacidic) and a strong base : $[\text{OH}^-]_f = C_1$ (just not considering the H^+ ions from WB).

- B-1.** The dissociation constant of acetic acid at a given temperature is 1.69×10^{-5} . The degree of dissociation of 0.01 M acetic acid in the presence of 0.01 M HCl is :
 (A) 1.69×10^{-7} (B) 1.69×10^{-5} (C) 1.69×10^{-3} (D) 2.9×10^{-2}
- B-2.** On adding 100 mL of 10^{-2} M NaOH solution to 100 mL of 0.01 M Triethyl amine solution ($K_b = 6.4 \times 10^{-5}$), change in pH of solution with respect to triethylamine solution will be :
 (A) - 0.8 (B) + 0.8 (C) + 1.1 (D) - 1.1
- B-3.** In above question, concentration of Triethyl ammonium ion ($[\text{C}_6\text{NH}_{16}^+]$) in resulting solution will be :
 (A) $100 K_b$ (B) $200 K_b$ (C) $10 K_b$ (D) K_b

Section (C) : pH calculation : Solutions containing mixture of weak acids, Solutions containing mixture of weak bases

Commit to memory :

Mixture of two weak acids (both monoprotic) : $[\text{H}^+] = C_1\alpha_1 + C_2\alpha_2 = \sqrt{C_1K_{a_1} + C_2K_{a_2}}$

Mixture of two weak bases (both monoacidic) : $[\text{OH}^-] = C_1\alpha_1 + C_2\alpha_2 = \sqrt{C_1K_{b_1} + C_2K_{b_2}}$

- C-1.** Consider an aqueous solution, 0.1 M each in HOCN, HCOOH, $(\text{COOH})_2$ and H_3PO_4 . For HOCN, we can write : $K_a(\text{HOCN}) = \frac{[\text{H}^+][\text{OCN}^-]}{[\text{HOCN}]}$. $[\text{H}^+]$ in this equation is :
 (A) H^+ ions released by HOCN
 (B) Sum of H^+ ions released by all monoprotic acids
 (C) Sum of H^+ ions released only the first dissociation of all the acids.
 (D) Overall H^+ ion concentration in the solution.
- C-2.** What are $[\text{H}^+]$, $[\text{A}^-]$ and $[\text{B}^-]$ in a solution that contains 0.03 M HA and 0.1 M HB. K_a for HA and HB are 3.0×10^{-4} and 1.0×10^{-10} respectively.
 (A) $[\text{H}^+] = 3 \times 10^{-3}$ M (B) $[\text{A}^-] = 3 \times 10^{-3}$ M
 (C) $[\text{B}^-] = 3.33 \times 10^{-9}$ M (D) All of these
- C-3.** pH of a solution obtained by mixing equal volumes of 0.1 M Triethyl amine ($K_b = 6.4 \times 10^{-5}$) & $\frac{4}{45}$ M NH_4OH ($K_b = 1.8 \times 10^{-5}$) will be :
 (A) 11.3 (B) 10.3 (C) 12.3 (D) 11.45
- C-4.** At 25°C, K_b for BOH = 1.0×10^{-12} . A 0.01 M solution of BOH has what value of $[\text{OH}^-]$?
 (A) 1.0×10^{-6} M (B) $\sqrt{2} \times 10^{-7}$ M (C) 1.0×10^{-5} M (D) 2.0×10^{-6} M


Section (D) : pH calculation : Solutions containing mixture of weak polyprotic acid and strong acid, Solutions containing mixture of weak Polyacidic base and strong base
Commit to memory :

 Mixture of a polyprotic weak acid and a strong acid : $[H^+] = [H^+]$ from SA

 Mixture of a polyprotic weak base and a strong base : $[OH^-] = [OH^-]$ from SB

D-1. For a solution obtained by mixing equal volumes of 0.02 M KOH solution & 0.2 M B (weak diacidic base; $K_{b1} = 10^{-7}$ & $K_{b2} = 10^{-14}$) solution :

- (A) pH = 12 (B)
- $[BH^+] = 10^{-6}$
- M (C)
- $[BH_2^{2+}] = 10^{-18}$
- M (D) All of these

Section (E) : Salt hydrolysis, pH calculation : Solutions of salt of monoprotic acid and monoacidic base.
Commit to memory :

 Salt of strong acid and weak base : $K_h \times K_b = K_w$; $h = \sqrt{\frac{K_h}{c}}$; $pH = \frac{1}{2} [pK_w - pK_b - \log c]$ (valid if $h < 0.1$ or 10%)

 Salt of strong base and weak acid : $K_h \times K_a = K_w$; $h = \sqrt{\frac{K_h}{c}}$; $pH = \frac{1}{2} [pK_w + pK_a + \log c]$ (valid if $h < 0.1$ or 10%)

 Salt of weak acid and weak base : $K_h \times K_a \times K_b = K_w$; $\left(\frac{h}{1-h}\right) = \sqrt{K_h}$; $pH = \frac{1}{2} [pK_w + pK_a - pK_b]$
E-1. The pH of a solution obtained by mixing 100 mL of 0.2 M CH_3COOH with 100 mL of 0.2 M NaOH would be : (pK_a for $CH_3COOH = 4.74$)

- (A) 4.74 (B) 8.87 (C) 9.10 (D) 8.57

Section (F) : pH calculation : Solutions of salt of weak polyprotic acid and strong base, Solutions of salt of weak Polyacidic base and strong acid, Solutions of amphiprotic species.
Commit to memory :

 Solution containing polyvalent anion (A^{x-}) of weak polyprotic acid (H_xA) :

$$K_{h1} = \frac{K_w}{K_{a_x}}; h = \sqrt{\frac{K_{h1}}{C}}; pH = \frac{1}{2} [pK_w + pK_{a_x} + \log C]$$

 Solution containing polyvalent cation (BH_y^{y+}) of weak polyprotic base (B) :

$$K_{h1} = \frac{K_w}{K_{b_y}}; h = \sqrt{\frac{K_{h1}}{C}}; pH = \frac{1}{2} [pK_w - pK_{b_y} - \log C]$$

 Solution containing amphiprotic anion of acid H_xA :

$$pH(H_{x-1}A^-) = \left(\frac{pK_{a1} + pK_{a2}}{2}\right); pH(H_{x-2}A^{2-}) = \left(\frac{pK_{a2} + pK_{a3}}{2}\right)$$

F-1. For a BH_2Cl_2 solution (chloride salt of a diacidic base B): (Dissociation constants of base are K_{b1} & K_{b2})

- (A)
- $K_{h1} \ll K_{h2}$
- (B)
- $K_{h1} = \frac{K_w}{K_{b1}}$
- (C)
- $K_{h1} = \frac{K_w}{K_{b2}}$
- (D)
- $K_{h2} = \frac{K_{b1}}{K_w}$

F-2. Select the correct statement regarding above solution :

- (A) Anion will undergo hydrolysis producing
- OH^-
- & solution is expected to be basic.
-
- (B) Anion will not undergo hydrolysis & solution is expected to be basic.
-
- (C) Cation will undergo hydrolysis producing
- H_3O^+
- & solution is expected to be acidic.
-
- (D) Cation will undergo hydrolysis producing
- OH^-
- & solution is expected to be acidic.



- F-3.** pH of 0.1 M Na_2HPO_4 and 0.2 M NaH_2PO_4 are respectively: (pK_a for H_3PO_4 are 2.2, 7.2 and 12)
 (A) 4.7, 9.6 (B) 9.6, 4.7 (C) 9.3, 4.4 (D) 4.4, 9.3
- F-4.** The pH of which salt solution is independent of its concentration ?
 1. $(\text{CH}_3\text{COO})\text{C}_5\text{H}_5\text{NH}$ 2. NaH_2PO_4 3. Na_2HPO_4 4. NH_4CN
 (A) 1, 2, 3, 4 (B) 1, 4 (C) 2, 3 (D) 1, 2, 3

Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

- In a solution obtained by mixing 100 mL of 0.25 M Triethylamine ($K_b = 6.4 \times 10^{-5}$) & 400 mL of $\frac{M}{18} \text{NH}_4\text{OH}$ ($K_b = 1.8 \times 10^{-5}$) :
 (A) $[\text{NH}_4^+] = 4 \times 10^{-4} \text{ M}$ (B) $[\text{C}_6\text{NH}_{16}^+] = 1.6 \times 10^{-3} \text{ M}$
 (C) Both (A) & (B) (D) None of these
- Ratio of $[\text{HA}^{2-}]$ in 1 L of 0.1 M H_3A solution ($K_{a_1} = 10^{-5}$; $K_{a_2} = 10^{-8}$ & $K_{a_3} = 10^{-11}$) & upon addition of 0.1 mole HCl to it will be :
 (A) 10 (B) 100 (C) 1000 (D) 10,000
- Calculate the degree of hydrolysis and pH of 0.005 M K_2CrO_4 . $K_{a_2} = 5 \times 10^{-7}$ for H_2CrO_4 . (It is essentially strong for first ionization)
 (A) $h = 2 \times 10^{-3}$ (B) $h = 0.01$ (C) pH = 5 (D) pH = 9.7

PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

- For a solution of weak triprotic acid H_3A ($K_{a_1} \gg K_{a_2}$, K_{a_3} ; $K_{a_2} = 10^{-8}$; $K_{a_3} = 10^{-13}$), $[\text{A}^{3-}] = 10^{-17} \text{ M}$. Determine pH of solution. Report your answer as '0', if you find data insufficient.
- Calculate $[\text{H}^+]$ in a 0.2 M solution of dichloroacetic acid ($K_a = 5 \times 10^{-2}$) that also contains 0.05 M H_2SO_4 . Report your answer after multiplying it by 400.
- Calculate the ratio of concentrations of HCOO^- & OCN^- ions in a solution containing 0.1 M HCOOH ($K_a = 1.8 \times 10^{-4}$) and 0.1 M HOCN ($K_a = 4 \times 10^{-4}$). If simplest ratio is a : b, report your answer as (a + b).
- In a solution containing 0.01 M HCl and 0.1 M H_2CO_3 , ratio of $[\text{H}^+]$ produced from strong acid & weak acid respectively is: 1. Report your answer as x/500. Given: K_{a_1} & K_{a_2} of H_2CO_3 are 4×10^{-7} & 4×10^{-11} respectively.
- Calculate the change in pH of 0.1 M Na_2HA solution after diluting it to ten times the original volume. (Given that for H_3A : $K_{a_1} = 10^{-4}$, $K_{a_2} = 10^{-7}$, $K_{a_3} = 10^{-11}$)
- What is the pOH of a 0.5 M Na_3PO_4 solution ? Report your answer as $10 \times \text{pOH}$. ($K_{a_1} = 7.5 \times 10^{-3}$, $K_{a_2} = 6.2 \times 10^{-8}$, $K_{a_3} = 4 \times 10^{-13}$)

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

- If K_{a_1} and K_{a_2} of H_2SO_4 are 10^{-2} and 10^{-6} respectively at a certain temperature, then :
 (A) $K_{a_1} > K_{a_2}$ because it is easy to abstract H^+ from H_2SO_4 and less easy to abstract H^+ from HSO_4^- .
 (B) K_{a_1} and K_{a_2} may be measured in acetic acid.
 (C) K_{a_1} and K_{a_2} are measured in H_2O .
 (D) the H^+ ion conc. of 0.01 M H_2SO_4 will be less than 0.02 M.



2. When 0.1 mole solid NaOH is added in 1 L of 0.1 M $\text{NH}_3(\text{aq})$ ($K_b = 2 \times 10^{-5}$), then select the correct statement(s) :
 (A) degree of dissociation of NH_3 approaches to zero.
 (B) change in pH by adding NaOH would be 1.85.
 (C) In solution, $[\text{Na}^+] = 0.1 \text{ M}$, $[\text{NH}_3] = 0.1 \text{ M}$, $[\text{OH}^-] = 0.2 \text{ M}$.
 (D) on addition of OH^- , K_b of NH_3 does not changes.
3. Which of the following solutions when added to 1L of a 0.01 M CH_3COOH solution will cause no change in the degree of dissociation of CH_3COOH and pH of the solution ? $K_a = 1.6 \times 10^{-5}$ for CH_3COOH ?
 (A) 0.6 mM HCOOH ($K_a = 8 \times 10^{-4}$) (B) 0.1 M CH_3COONa
 (C) 0.4 mM HCl (D) 0.01 M CH_3COOH

Exercise-3

* Marked questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

1. Will the pH of water be same at 4°C and 25°C ? Explain. [JEE-2003, 2/60]
2. A weak acid HX at 25°C has the dissociation constant 1×10^{-5} . It forms a salt NaX on reaction with caustic soda. The percentage hydrolysis of 0.1 M solution of NaX is : [JEE-2004(S), 3/84]
 (A) 0.0001% (B) 0.01 % (C) 0.1 % (D) 0.15 %
3. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4} . The pH of 0.01 M solution of its sodium salt is : [JEE-2009, 4/160]
4. The total number of diprotic acids among the following is : [JEE 2010, 3/163]
 H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $\text{H}_2\text{S}_2\text{O}_7$
 H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3
5. Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is : [JEE-2010, 3/163]
 KCN K_2SO_4 $(\text{NH}_4)_2\text{C}_2\text{O}_4$ NaCl $\text{Zn}(\text{NO}_3)_2$
 FeCl_3 K_2CO_3 NH_4NO_3 LiCN

PART - II : JEE (MAIN) (PREVIOUS YEARS)

JEE(MAIN) OFFLINE PROBLEMS

1. In aqueous solution, the ionization constants for carbonic acid are : [AIEEE-2010, 4/144]
 $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$
 Select the correct statement for a saturated 0.034 M solution of the carbonic acid.
 (1) The concentration of CO_3^{2-} is 0.034 M.
 (2) The concentration of CO_3^{2-} is greater than that of HCO_3^- .
 (3) The concentration of H^+ and HCO_3^- are approximately equal.
 (4) The concentration of H^+ is double that of CO_3^{2-} .
2. An aqueous solution contains 0.10 M H_2S and 0.20 M HCl . If the equilibrium constant for the formation of HS^- from H_2S is 1.0×10^{-7} and that of S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is : [JEE(Main)-2018, 4/120]
 (1) 6×10^{-21} (2) 5×10^{-19} (3) 5×10^{-8} (4) 3×10^{-20}



Answers

EXERCISE - 1

PART - I

- A-1.** $\text{pH} = 4.52$, $3 \times 10^{-5} \text{ M}$, $1.2 \times 10^{-13} \text{ M}$ **A-2.** 10.91
B-1. -1.52 , $9 \times 10^{-5} \text{ M}$. **B-2.** 12 , $2 \times 10^{-4} \text{ M}$, 0.2%
C-1. $[\text{H}^+] = 8 \times 10^{-3} \text{ M}$, $[\text{HCOO}^-] = 3 \times 10^{-3} \text{ M}$, $[\text{OCN}^-] = 5 \times 10^{-3} \text{ M}$
D-1. 10^{-6} M , 10^{-18} M **E-1.** 11 **E-2.** $\text{pH} = 4.19$

PART - II

- | | | |
|-----------------|-----------------|---|
| A-1. (C) | A-2. (C) | A-3. (i) (B) (ii) (A) (iii) (A) (iv) (B) (v) (B) |
| A-4. (C) | B-1. (C) | B-2. (B) B-3. (D) C-1. (D) |
| C-2. (D) | C-3. (A) | C-4. (B) D-1. (D) E-1. (B) |
| F-1. (C) | F-2. (C) | F-3. (B) F-4. (A) |

EXERCISE - 2

PART - I

1. (C) 2. (D) 3. (A)

PART - II

1. 4 2. 60 3. 29 (actual answer = 9 : 20) 4. 5
 5. 0 6. 10

PART - III

1. (ABD) 2. (ABD) 3. (ACD)

EXERCISE - 3

PART - I

1. It will not be same at two different temperatures.
 2. (B) 3. 8 4. 6 5. 3

PART - II

JEE(MAIN) OFFLINE PROBLEMS

1. (3) 2. (4)